Engineering Materials

Chapter 2: Atomic Structure and Interatomic Bonding



Objectives

- Name the two atomic models cited, and note the differences between them.
- Describe the important quantum-mechanical principle that relates to electron energies.
- Briefly describe ionic, covalent, metallic, hydrogen, and van der Waals bonds. Note which materials exhibit each of these bonding types.



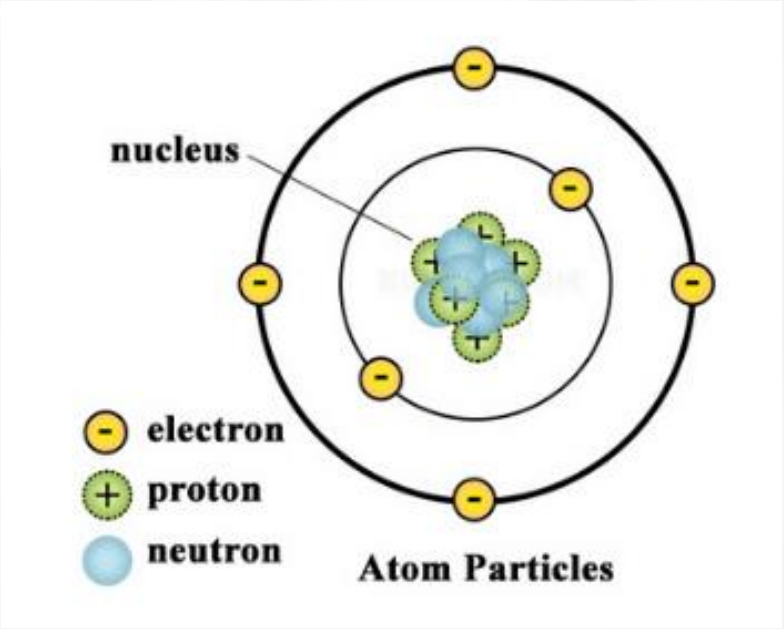
Importance of the topic

- Type of bond allows us to explain a material's properties. For example, carbon, may exist as both graphite and diamond due to the type of interatomic bonding found in them.
- The type of bonding has an influence on the structures of the material.



Atomic structure

- Each atom consists of a very small nucleus composed of protons and neutrons, which is encircled by moving electrons.
- Both electrons and protons are electrically charged (Negative and positive charges respectively of magnitude 1.6×10^{-19} Coulombs)
- Neutrons are electrically neutral.
- Each chemical element is characterized by the number of protons in the nucleus, or the ***atomic number (Z)***



Atomic structure

- The ***atomic mass (A)*** of a specific atom may be expressed as the sum of the masses of protons and neutrons within the nucleus.
- Atoms of some elements have two or more different atomic masses, which are called ***isotopes.***
- The ***atomic weight*** of an element corresponds to the weighted average of the atomic masses of the atom's naturally occurring isotopes.



Atomic Weight

- The atomic weight of an element or the molecular weight of a compound may be specified on the basis of amu per atom (molecule) or mass per mole of material.
- In one ***mole*** of a substance there are 6.022×10^{23} (Avogadro's number) atoms or molecules.
- These two atomic weight schemes are related through the following equation:

  1 amu/atom (or molecule) = 1 g/mol

- For example, the atomic weight of iron is 55.85 amu/atom, or 55.85 g/mol.
- Sometimes use of amu per atom or molecule is convenient; on other occasions grams (or kilograms) per mole is preferred.



Electrons in Atoms – Bohr atomic model

- Electrons revolve around the atomic nucleus in discrete orbitals.
- Position of an electron is well defined in terms of its orbital
- The Bohr model represents an early attempt to describe electrons in atoms, in terms of both position (electron orbitals) and energy.

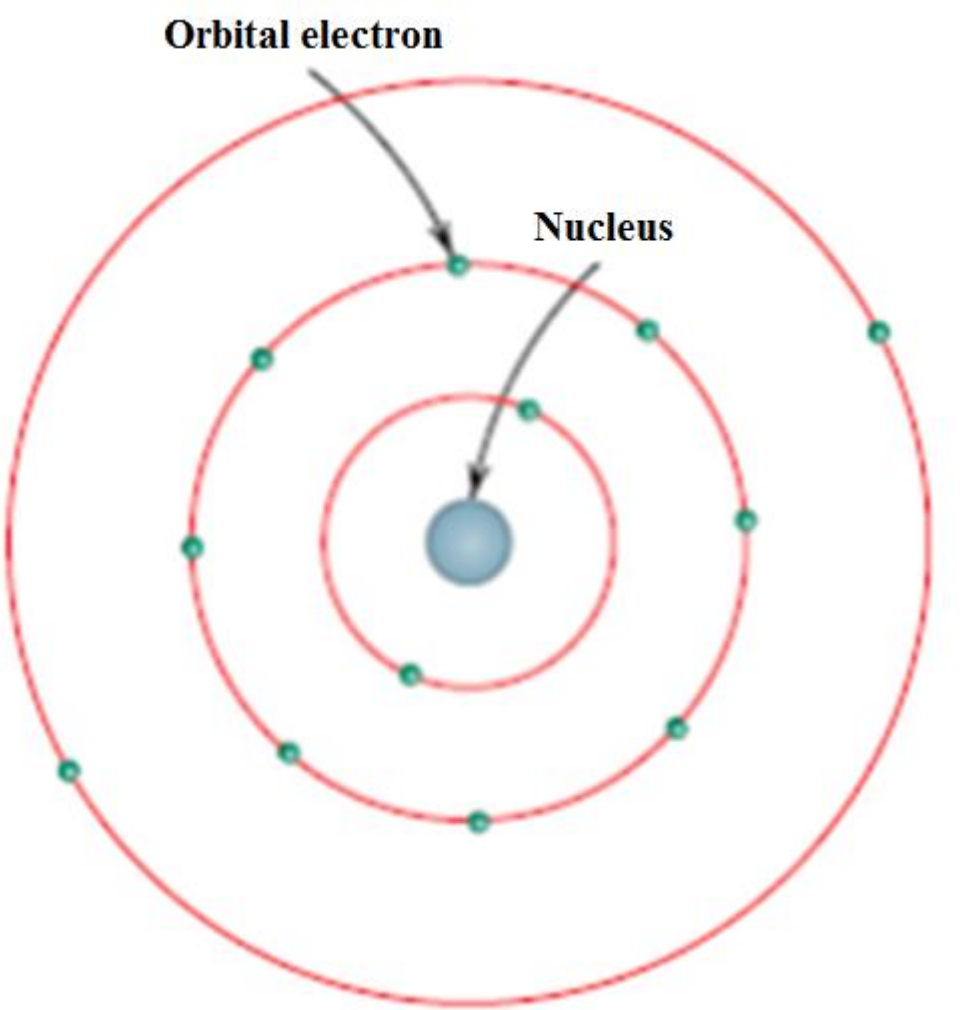


Fig. Schematic representation of Bohr atom

Wave-mechanical model

- The electron is considered to exhibit both wavelike and particle-like characteristics.
- With this model, an electron is no longer treated as a particle moving in a discrete orbital; rather,
- Position of an electron is described by a probability distribution or electron cloud.

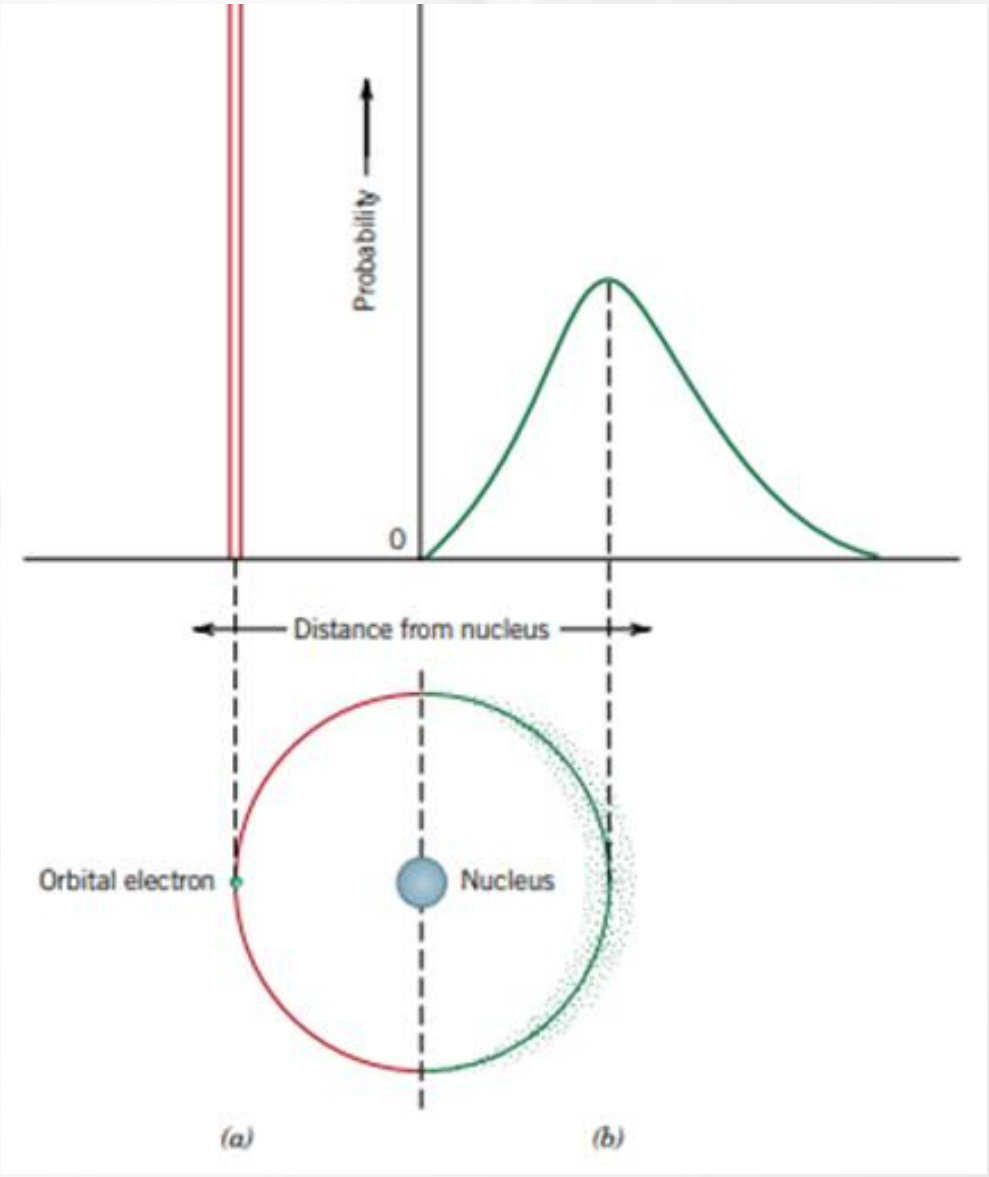


Fig. (a) Bohr and (b) wave-mechanical atom models

Quantum Numbers

- Using wave mechanics, every electron in an atom is characterized by four parameters called quantum numbers.
- The size, shape, and spatial orientation of an electron's probability density are specified by three of these quantum numbers.
- Bohr energy levels separate into electron subshells, and quantum numbers dictate the number of states within each subshell.



Quantum Numbers

The number of available states in electron shells & subshells

Principal Q. N., n		Subshells	Number of States	Number of Electrons Per Subshell	Number of Electrons Per Shell
1 (l=0)	*K-shell*	s	1	2	2
2 (l=0)	*L-shell*	s	1	2	8
2 (l=1)		p	3	6	
3 (l=0)	*M-shell*	s	1	2	18
3 (l=1)		p	3	6	
3 (l=2)		d	5	10	
4 (l=0)	*N-shell*	s	1	2	32
4 (l=1)		p	3	6	
4 (l=2)		d	5	10	
4 (l=3)		f	7	14	



Quantum Numbers

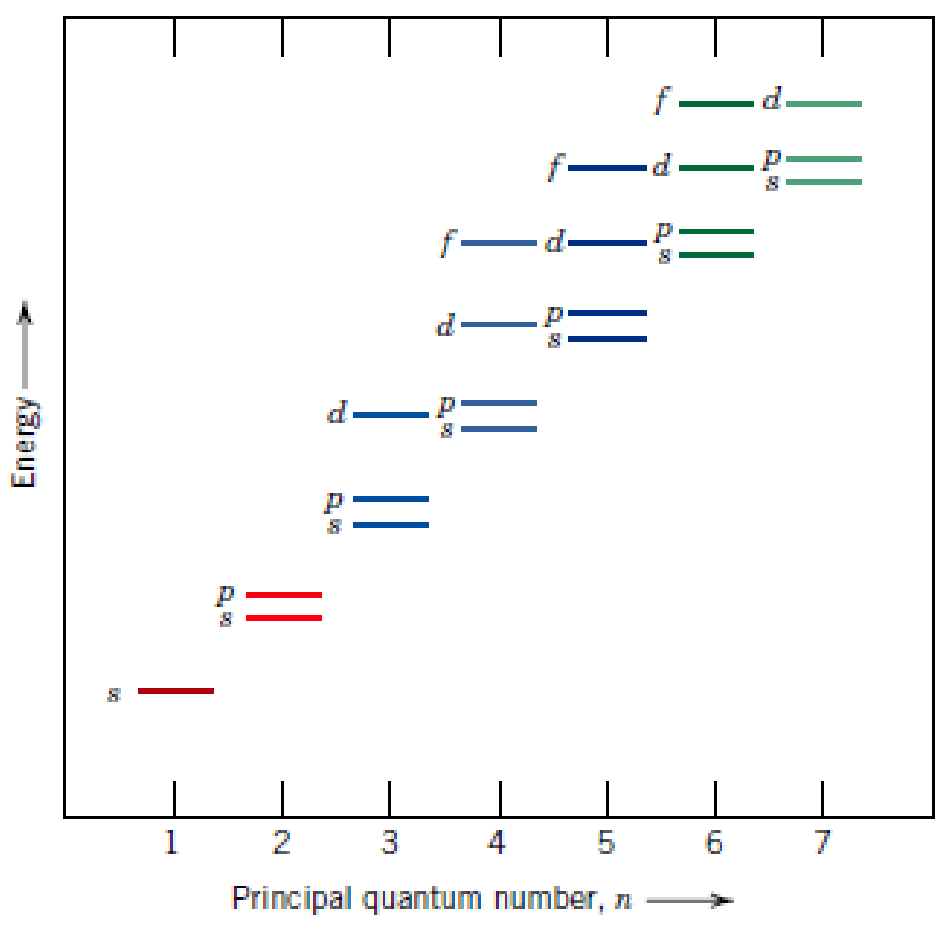


Fig. Schematic representation of the relative energies of the electrons for the various shells and subshells

Electron Configurations

- The ***Pauli exclusion principle*** is used to determine the manner in which states of atoms are filled with electrons.
- This principle stipulates that each electron state can hold no more than two electrons, which must have opposite spins. Thus, *s*, *p*, *d*, and *f* subshells may each accommodate, respectively, a total of 2, 6, 10, and 14 electrons.



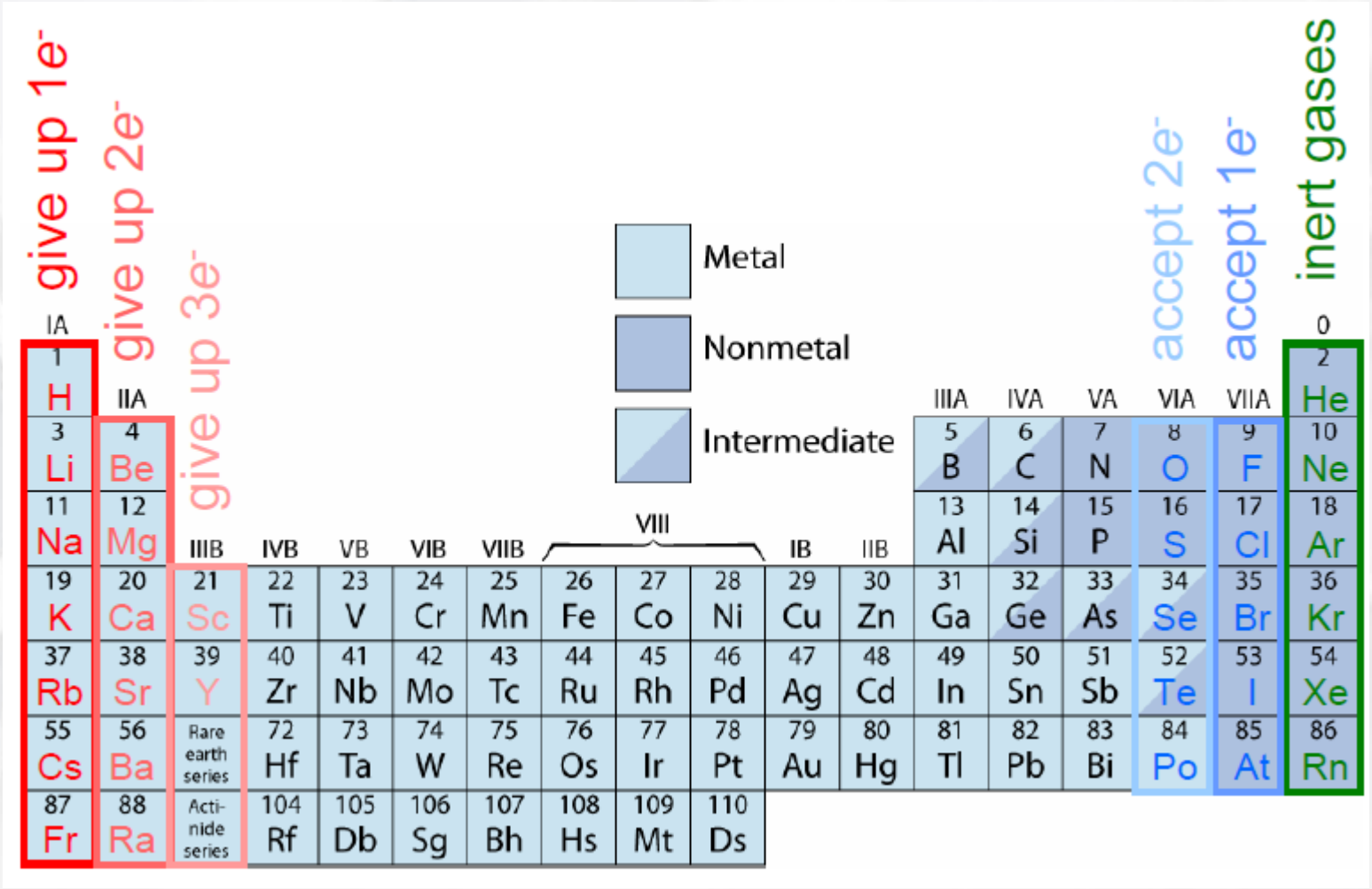
Electron Configurations

Element	Atomic #	Electron configuration	
Hydrogen	1	$1s^1$	
Helium	2	$1s^2$	(stable)
Lithium	3	$1s^2 2s^1$	
Beryllium	4	$1s^2 2s^2$	
Boron	5	$1s^2 2s^2 2p^1$	
Carbon	6	$1s^2 2s^2 2p^2$	
...		...	
Neon	10	$1s^2 2s^2 2p^6$	(stable)
Sodium	11	$1s^2 2s^2 2p^6 3s^1$	
Magnesium	12	$1s^2 2s^2 2p^6 3s^2$	
Aluminum	13	$1s^2 2s^2 2p^6 3s^2 3p^1$	
...		...	
Argon	18	$1s^2 2s^2 2p^6 3s^2 3p^6$	(stable)
...	...	...	
Krypton	36	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$	(stable)

Electron configurations where all states within valence electron shell are filled are stable unreactive inert or noble gas.

The Periodic Table

- **Columns: Similar Valence Structure**



Electropositive elements: Readily give up electrons to become + ions.

Electronegative elements: Readily acquire electrons to become - ions.

Periodic Table - Electronegativity

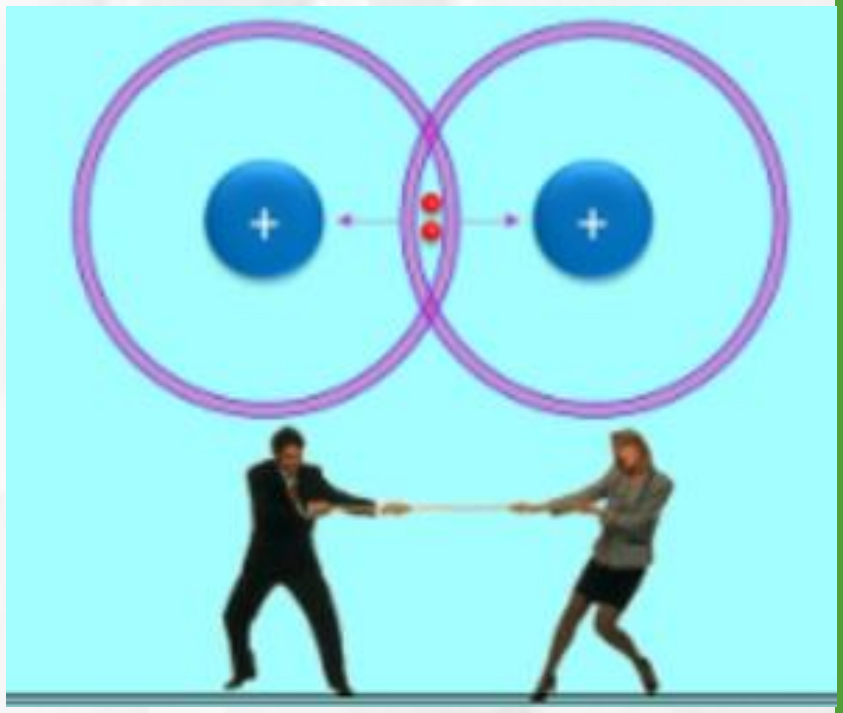
IA																	0
1 H 2.1	IIA											IIIA	IVA	VA	VIA	VIIA	2 He –
3 Li 1.0	4 Be 1.5											5 B 2.0	6 C 2.5	7 N 3.0	8 O 3.5	9 F 4.0	10 Ne –
11 Na 0.9	12 Mg 1.2	IIIB	IVB	VB	VIB	VIIB	VIII			IB	IIB	13 Al 1.5	14 Si 1.8	15 P 2.1	16 S 2.5	17 Cl 3.0	18 Ar –
19 K 0.8	20 Ca 1.0	21 Sc 1.3	22 Ti 1.5	23 V 1.6	24 Cr 1.6	25 Mn 1.5	26 Fe 1.8	27 Co 1.8	28 Ni 1.8	29 Cu 1.9	30 Zn 1.6	31 Ga 1.6	32 Ge 1.8	33 As 2.0	34 Se 2.4	35 Br 2.8	36 Kr –
37 Rb 0.8	38 Sr 1.0	39 Y 1.2	40 Zr 1.4	41 Nb 1.6	42 Mo 1.8	43 Tc 1.9	44 Ru 2.2	45 Rh 2.2	46 Pd 2.2	47 Ag 1.9	48 Cd 1.7	49 In 1.7	50 Sn 1.8	51 Sb 1.9	52 Te 2.1	53 I 2.5	54 Xe –
55 Cs 0.7	56 Ba 0.9	57–71 La–Lu 1.1–1.2	72 Hf 1.3	73 Ta 1.5	74 W 1.7	75 Re 1.9	76 Os 2.2	77 Ir 2.2	78 Pt 2.2	79 Au 2.4	80 Hg 1.9	81 Tl 1.8	82 Pb 1.8	83 Bi 1.9	84 Po 2.0	85 At 2.2	86 Rn –
87 Fr 0.7	88 Ra 0.9	89–102 Ac–No 1.1–1.7															

Smaller electronegativity

Larger electronegativity

Electronegativity

- The ability of an atom to attract the shared pair of electrons towards itself in a molecule, is called ***Electronegativity***.
- Electronegativity increases from left to right on the Periodic Table.
- Subshells with one electron have low electronegativity.
- Subshells with one missing electron have high electronegativity.
- Metals are electropositive because they can give up their few valence electrons to become positively charged ions.

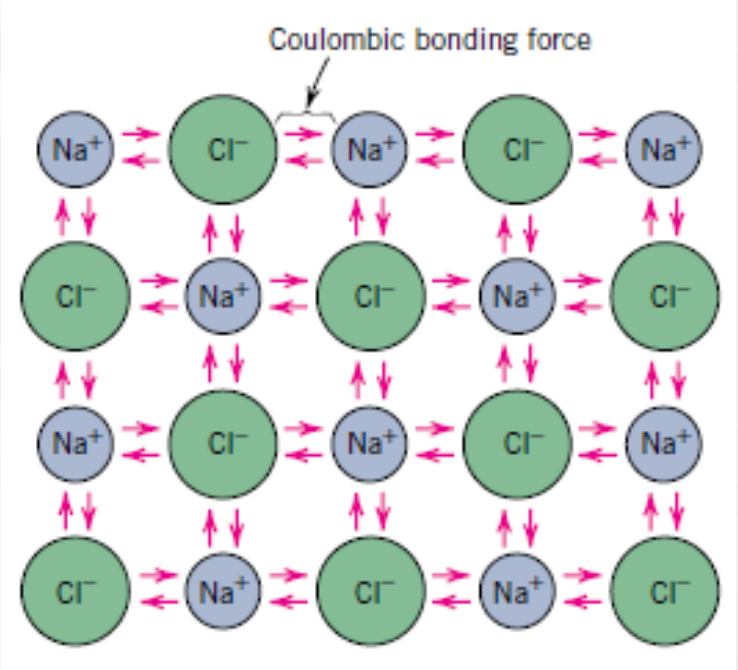


Think-pair-share (15 Mins)

1. How does the energy levels of Helium and Calcium affect their reactivity? Which of the two atoms will be much more reactive?
2. Give the electronic configuration of the following ions
   - Fe^{2+}
   - Al^{3+}
   - O^{2-}

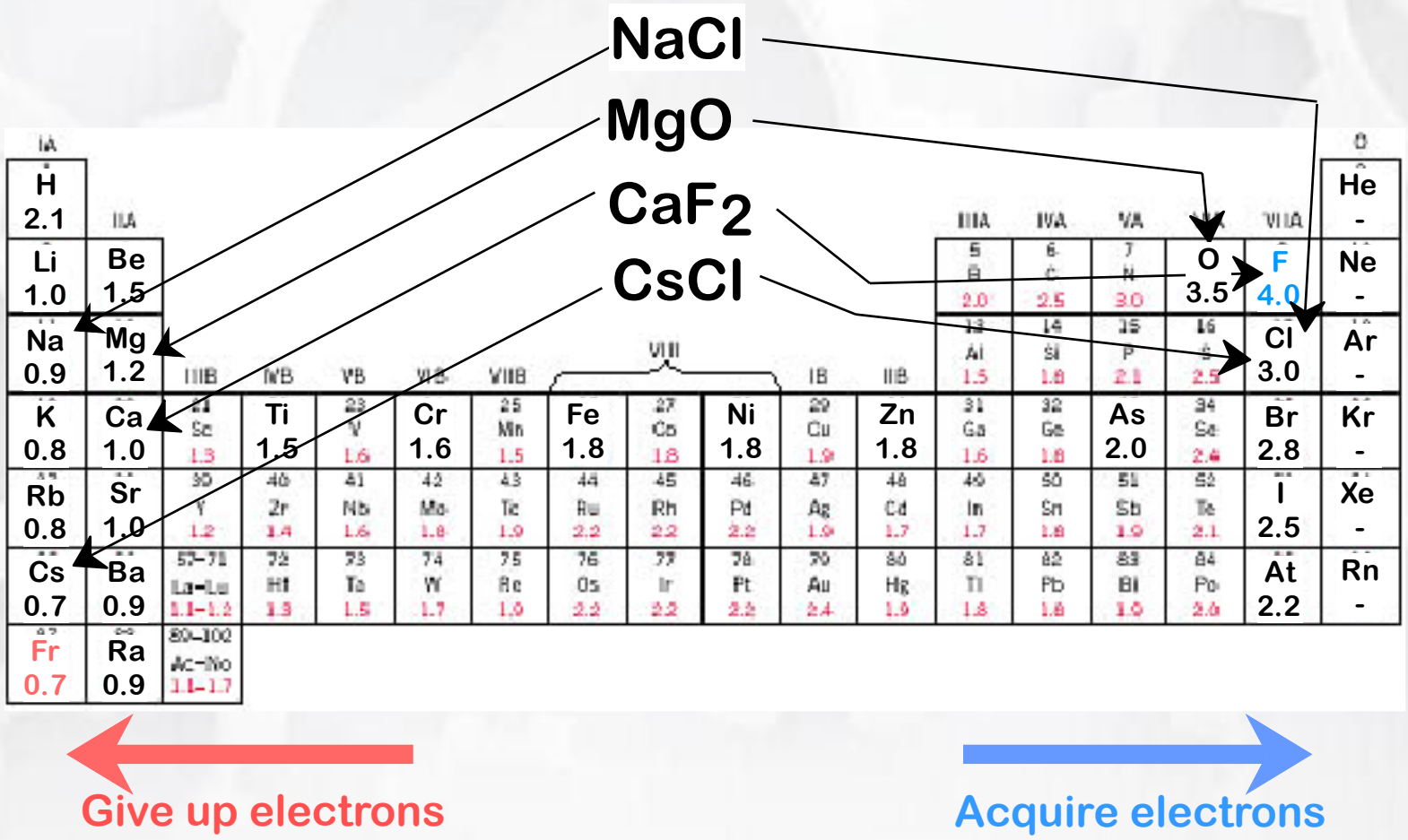
Atomic bonding in solids – Ionic bonding

- **Ionic bonding** is found in compounds that are composed of both metallic and non-metallic elements (usually found at the horizontal extremities of the periodic table).
- Atoms of a metallic element easily ***donate*** their ***valence electrons*** to the non-metallic atoms.
- In the process all the atoms acquire ***stable configurations*** and become ***ions***.
- The attractive bonding forces are ***coulombic forces;*** that is, positive and negative ions, by virtue of their net electrical charge, attract one another.



Examples: Ionic Bonding

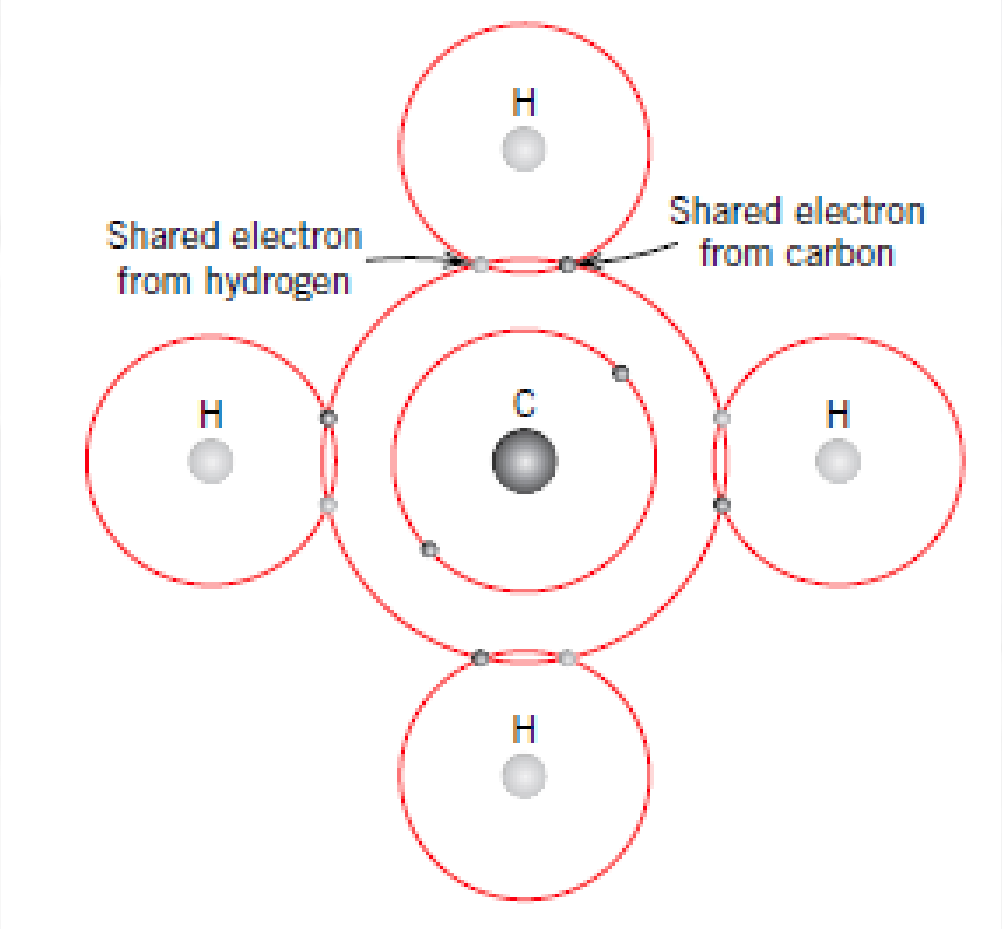
- Predominant bonding in **Ceramics**



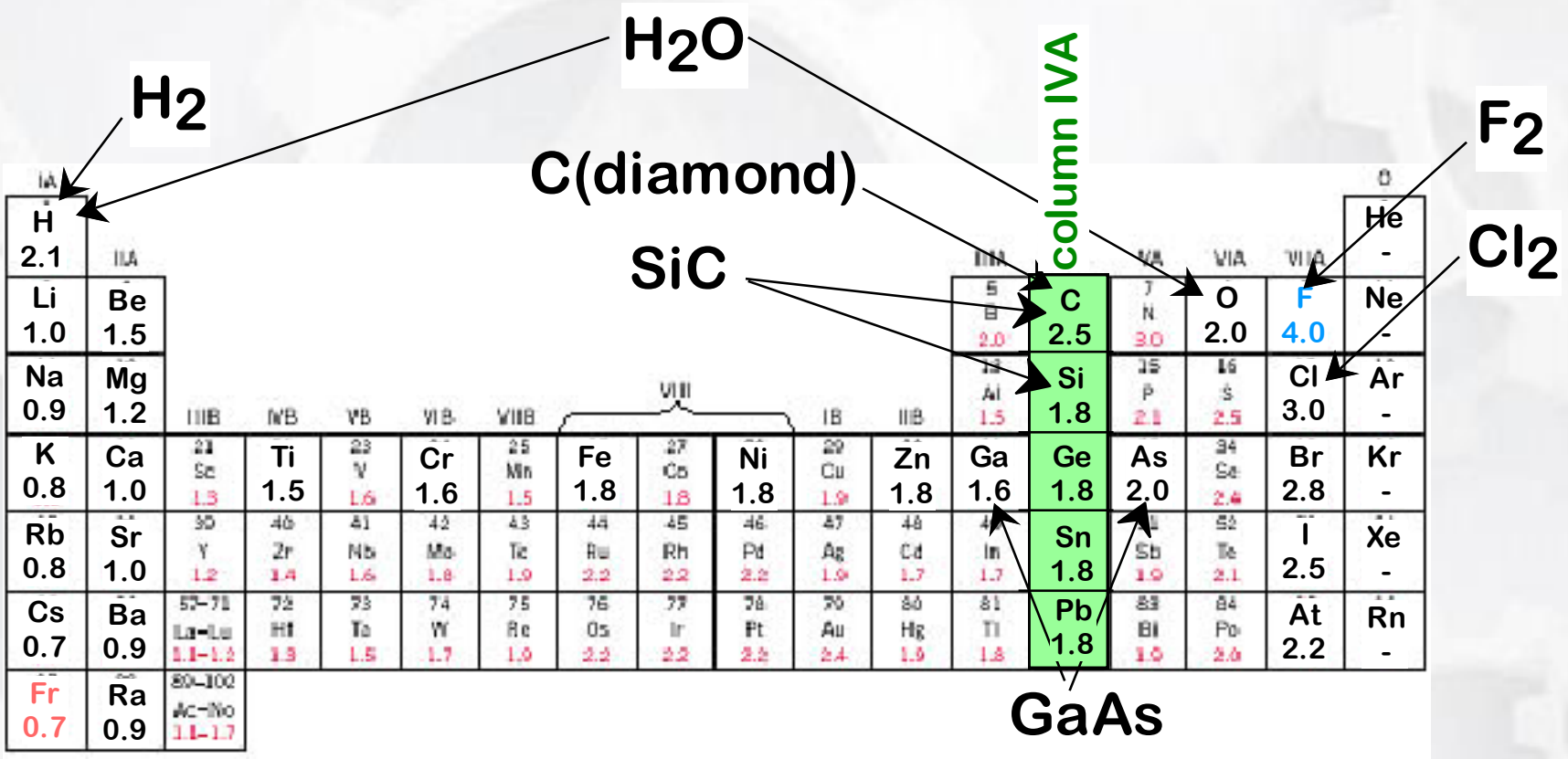
Adapted from Fig. 2.7, *Callister 6e.* (Fig. 2.7 is adapted from Linus Pauling, *The Nature of the Chemical Bond*, 3rd edition, Copyright 1939 and 1940, 3rd edition. Copyright 1960 by Cornell University.

Covalent bonding

- Stable electron configurations are assumed by the sharing of electrons between adjacent atoms.



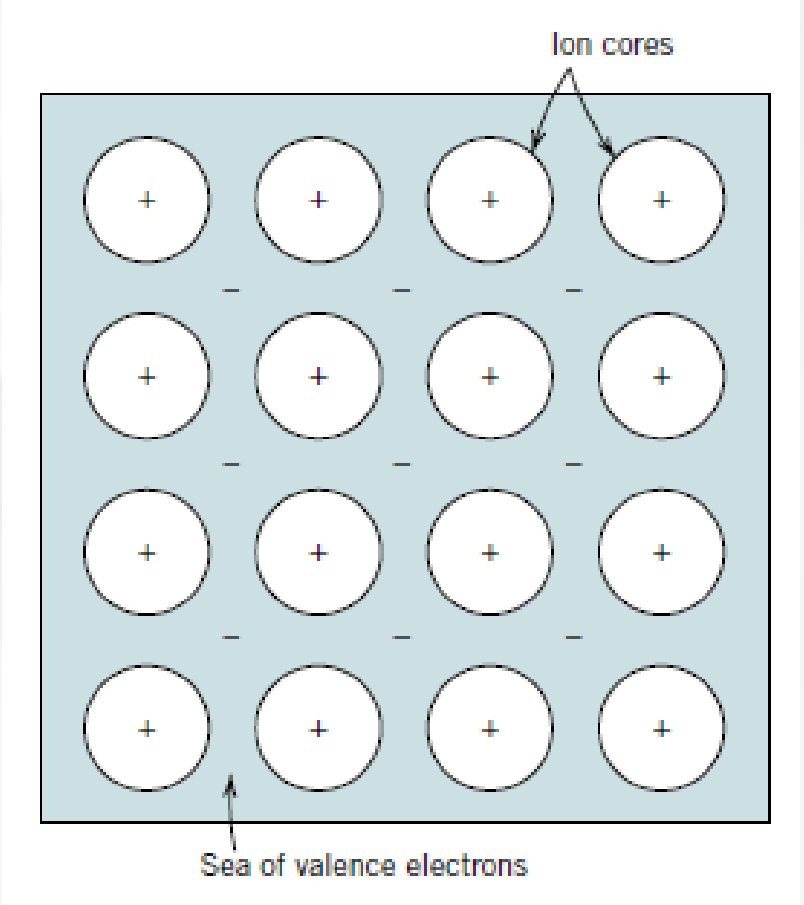
Examples: Covalent Bonding



- Molecules with nonmetals
- Molecules with metals and nonmetals
- Elemental solids (RHS of Periodic Table)
- Compound solids (about column IVA)

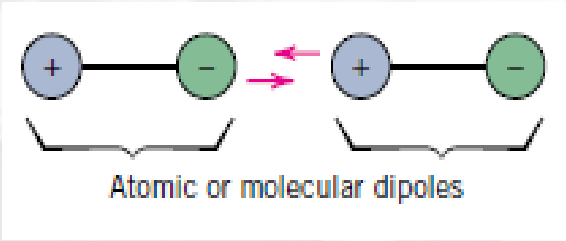
Metallic bonding

- Primary bond for metals and their alloys
- Arises from a sea of donated valence electrons (1, 2, or 3 from each atom).

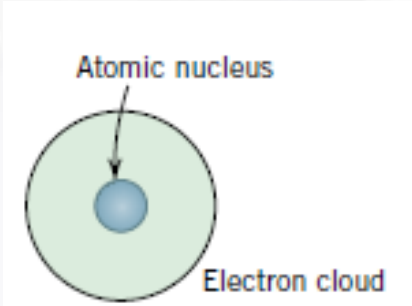


Secondary Bonding or Van Der Waals Bonding

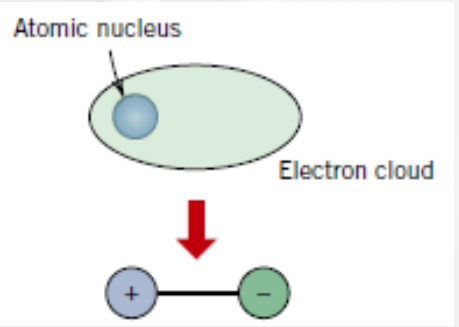
- Secondary bonding forces arise from atomic or molecular **dipoles.**



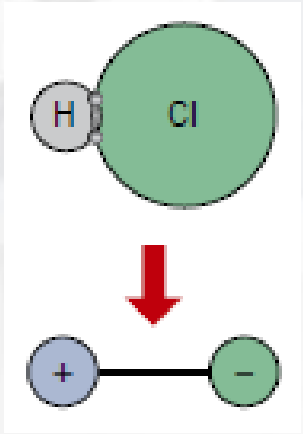
Van der Waals bonding between two dipoles.



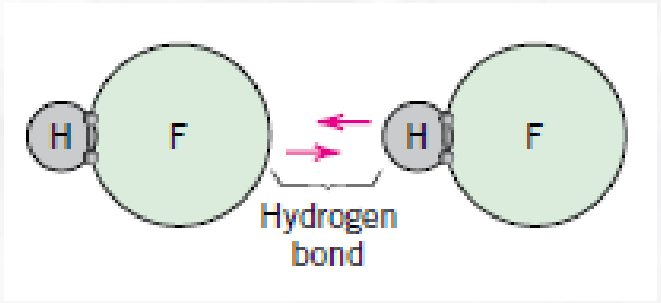
An electrically symmetric atom



An induced atomic dipole



A polar hydrogen chloride (HCl) molecule



Hydrogen bonding in hydrogen fluoride (HF).

Summary – Electrons in Atoms

- The two atomic models are Bohr and wave-mechanical.
- Whereas the Bohr model assumes electrons to be particles orbiting the nucleus in discrete paths, wave mechanics considers electrons to be wavelike and their positions are treated in terms of a probability distribution.
- The energies of electrons are quantized - that is, only specific values of energy are allowed.
- The four electron quantum numbers each specifies a distinct electron characteristic.
- According to the Pauli exclusion principle, each electron state can accommodate no more than two electrons, which must have opposite spins.



Summary – Periodic Table

- Elements in each of the columns (or groups) of the periodic table have distinctive electron configurations.
- For example, Group 0 elements (the inert gases) have filled electron shells, and Group IA elements (the alkali metals) have one electron greater than a filled electron shell.



Summary – Interatomic Bonding

- For ***ionic bonds***, electrically charged ions are formed by the transference of valence electrons from one atom type to another. This type of bonding is found in ceramic materials.
- There is a sharing of valence electrons between adjacent atoms when bonding is ***covalent.*** Polymers and some ceramic materials covalently bond.
- With ***metallic bonding***, the valence electrons form a “sea of electrons” that is uniformly dispersed around the metal ion cores and acts as a form of glue for them. Metallic materials exhibit this type of bonding



Summary – Secondary Bonding or van der Waals Bonding

- Relatively weak van der Waals bonds result from attractive forces between electric dipoles, which may be induced or permanent.
- For hydrogen bonding, highly polar molecules form when hydrogen covalently bonds to a non-metallic element such as fluorine.



Assignment

- Upon freezing (i.e., transforming from a liquid to a solid upon cooling), most substances experience an increase in density (or, correspondingly, a decrease in volume). One exception is water, which exhibits the anomalous and familiar expansion upon freezing - approximately 9 volume percent expansion. Please explain why water exhibits this behaviour on the basis of its interatomic bonding.